

Hierarchical Multi-Label Classification of Unlabeled Amazon Product Reviews

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Hierarchical multi-label text classification assigns multiple labels to documents when the label space is structured as a taxonomy. This task is particularly challenging without labeled training data. In this project, we study unsupervised hierarchical multi-label classification of Amazon product reviews, where each review is associated with two to three product categories organized in a directed acyclic graph.

We propose a pipeline that combines zero-shot semantic representations, similarity-based silver label generation, and hierarchy-aware graph neural networks (GNNs). The methodology used in this project is adapted from TaxoClass paper [1]. First, review texts and class descriptions augmented with class-related keyword contexts are encoded using pretrained sentence encoders. Silver labels are generated by mining confident core classes through a top-down traversal of the class taxonomy, supported by a synthetic root to handle forest-structured hierarchies. Finally, a GNN-based classifier propagates information throughout the class hierarchy to produce document-level predictions. Our best model achieves a precision of 0.42525 on the test documents.

Additional Key Words and Phrases: hierarchical multi-label classification, silver labels, graph neural networks, sentence embeddings

1 Introduction

Hierarchical multi-label text classification (HMTC) appears in settings where documents belong to multiple categories, where their relationships are organized in a taxonomy. Product categorization in e-commerce is a representative example: a single review often implies both specific leaf categories and their broader parent categories.

In this project, we address HMTC in an unsupervised setting with Amazon product reviews. We are given unlabeled train and test corpus, 531 product categories, their hierarchical relationships, and class-related keywords. Since no ground-truth labels are available for training, we rely on (i) similarity-based silver label generation and (ii) hierarchy-aware learning.

Our pipeline has three stages: (1) zero-shot encoding of reviews and class descriptions, (2) core class mining to produce reliable silver labels guided by the taxonomy, and (3) training a GNN classifier that models dependencies between classes.

2 Task Definition

Formally, the input consists of an unlabeled corpus D and a class taxonomy $T = (C, R)$, where each node $c \in C$ is a product class and each directed edge $(c_i, c_j) \in R$ denotes a parent-child relationship. The goal is to learn a classifier $f(\cdot)$ that maps a document $d \in D$ to a binary label vector $\mathbf{y} \in \{0, 1\}^{|C|}$. In this specific unlabeled corpus, each document is associated with at least two and at most three labels.

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Resources provided: (i) 29,487 unlabeled training reviews, (ii) 19,658 test reviews for evaluation, (iii) 531 classes, (iv) a class hierarchy taxonomy, and (v) class-related keywords.

3 Silver Label Generation

3.1 Zero-shot Encoding of Reviews and Classes

We encode review texts and class descriptions using a pretrained sentence transformer, `all-mpnet-base-v2`. Both test and train text corpus are encoded to be used during silver label generation and classifier training. To enrich class semantics, we combine each class name with its keywords and use an entailment-style prompt, e.g., *“This product category is {class}. Typical keywords include: {keywords}. Reviews in this category describe products related to {class}.”* Embeddings are computed in a zero-shot manner, L2-normalized, and cached for downstream use.

3.2 Similarity Ensemble

We compute cosine similarity between review embeddings and class embeddings to obtain dense semantic similarity. To complement this with lexical signals, we compute TF-IDF vectors over reviews and class texts and measure cosine similarity in sparse space. Each similarity matrix is min-max normalized, then combined:

$$\text{sim}_{\text{Ensemble}}(d, c) = \alpha \text{sim}_{\text{SBERT}}(d, c) + \beta \text{sim}_{\text{TF-IDF}}(d, c),$$

where $\alpha = 0.7$ and $\beta = 0.3$.

3.3 Core Class Mining with Synthetic Root

The taxonomy forms a forest where multiple nodes with in-degree of zero exists. To enable a unified top-down traversal and make sure that all classes are considered, we add a synthetic root connected to all original roots.

For each document, candidate classes are mined iteratively: starting from the synthetic root, we select the most similar children to expand. At depth l , for each queued class we select its top $(l + 2)$ children by similarity score. We then keep the top $(l + 1)^2$ candidates based on a path score (ps) that multiplies similarities along the traversal path defined as:

$$\begin{aligned} ps(\text{Root}) &= 1, \\ ps(c_j) &= \max_{c_k \in \text{Par}(c_j)} \{ps(c_k) \cdot \text{sim}(c_j, D)\}. \end{aligned}$$

3.4 Confident Core Identification

From the mined candidates, we identify confident core classes by comparing a class similarity against competitors (parents and siblings). Because similarity distributions were relatively flat, median-based filtering alone did not sufficiently prune candidates. We apply:

$$\tau = \max(\text{median confidence}, 0.04),$$

and keep candidate cores with confidence $\geq \tau$. We output two to three most confident core classes per document as silver labels.

4 Model and Training Process

4.1 Hierarchy-aware GNN Classifier

We train a GNN-based multi-label classifier over the class taxonomy. Let initial class embeddings be $H^{(0)} \in \mathbb{R}^{|C| \times d}$ initialized from sentence embeddings and treated as learnable parameters. We

build a global normalized adjacency matrix $\hat{A}$ by adding undirected parent-child edges, self-loops, and applying symmetric normalization $D^{-1/2}AD^{-1/2}$.

A two-layer GCN updates class representations:

$$H^{(l+1)} = \sigma(\hat{A}H^{(l)}W^{(l)}),$$

with ReLU and dropout between layers. Final class vectors are projected to the document space through a bilinear matrix B , and logits are computed via normalized dot-product between projected class vectors and document embeddings.

4.2 Training Objective and Imbalance Handling

We train using silver labels: positives include core classes and all ancestors; negatives exclude positives and descendants of core classes. A class imbalance arises due to the imbalance of positive and negative sets since the number of negative classes is significantly larger than the positive class. This causes the learning signals to be skewed towards negatives without learning anything valuable from the positives. To reduce imbalance, the number of negative classes is capped at 50 per document.

We optimize binary cross-entropy with logits loss using class-wise positive weights and AdamW with learning rate 4×10^{-3} . We use early stopping (patience 6) with a maximum of 200 epochs.

5 Prediction and Evaluation

Both training and test corpora are used to generate silver labels and train the classifier in unlabeled setting. Evaluation is performed on the test set only.

In inference, we compute logits for all classes and output the top-3 classes per document. A 2-vs-3 label heuristic was considered but not used in the final submission; the predictions are top-3 unconditionally.

6 Experimental Results

Table 1. Precision on the test set.

Method	Precision
SBERT embeddings + GNN (proposed)	0.42525
BERT-base embeddings + GNN	0.31965
Naïve fine-tuned BERT classifier	0.21148

Using sentence-level embeddings together with hierarchical GNN propagation yields the best performance. Replacing sentence embeddings with vanilla bert-base-uncased embeddings reduces precision, indicating the importance of stronger semantic encoders for similarity-based silver labeling and downstream training. A straightforward approach to finetuning the BERT classifier without hierarchy modeling also performed poorly compared to the proposed method.

7 Case Study and Discussion

We compare two semantically similar reviews about hair styling tools. Both belong to the broader styling tool category, but one review discusses a hair straightener while the other focuses on a hair dryer.

Straightener review. An excerpt from the review states: “*i bought it only because i was desperate for a cheap **straightner** because my **hair** is very thick , long , wavy.*” The model predicts classes 64:hair care, 43:shaving hair removal, 10:beauty. Misspellings such as **straightner** and repeated generic mentions of **hair** bias the similarity scores toward broader beauty and hair-care categories rather than the specific styling tool.

Dryer review. In contrast, the second review explicitly mentions: “*i was so excited to receive this **hairdryer** ... it overheats really quickly.*” Here, the model predicts 64:hair care, 93:styling tools, 10:beauty, correctly identifying the styling tools subcategory.

This analysis suggests that the pipeline is sensitive to explicit lexical product cues, while noisy writing and indirect references can cause confusion among sibling categories even with hierarchical structure.

8 Conclusion

We presented an unsupervised pipeline for hierarchical multi-label classification of product reviews. By combining zero-shot sentence embeddings, similarity-based core class mining with a synthetic root, and a hierarchy-aware GNN classifier, we achieve a best precision of 0.42525 on the test set and outperform simpler baselines. Future work could improve robustness to misspellings and indirect descriptions, and explore additional weak-supervision strategies.

References

- [1] Jiaming Shen, Wenda Qiu, Yu Meng, Jingbo Shang, Xiang Ren, and Jiawei Han. 2021. TaxoClass: Hierarchical Multi-Label Text Classification Using Only Class Names. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou (Eds.). Association for Computational Linguistics, Online, 4239–4249. doi:10.18653/v1/2021.naacl-main.335